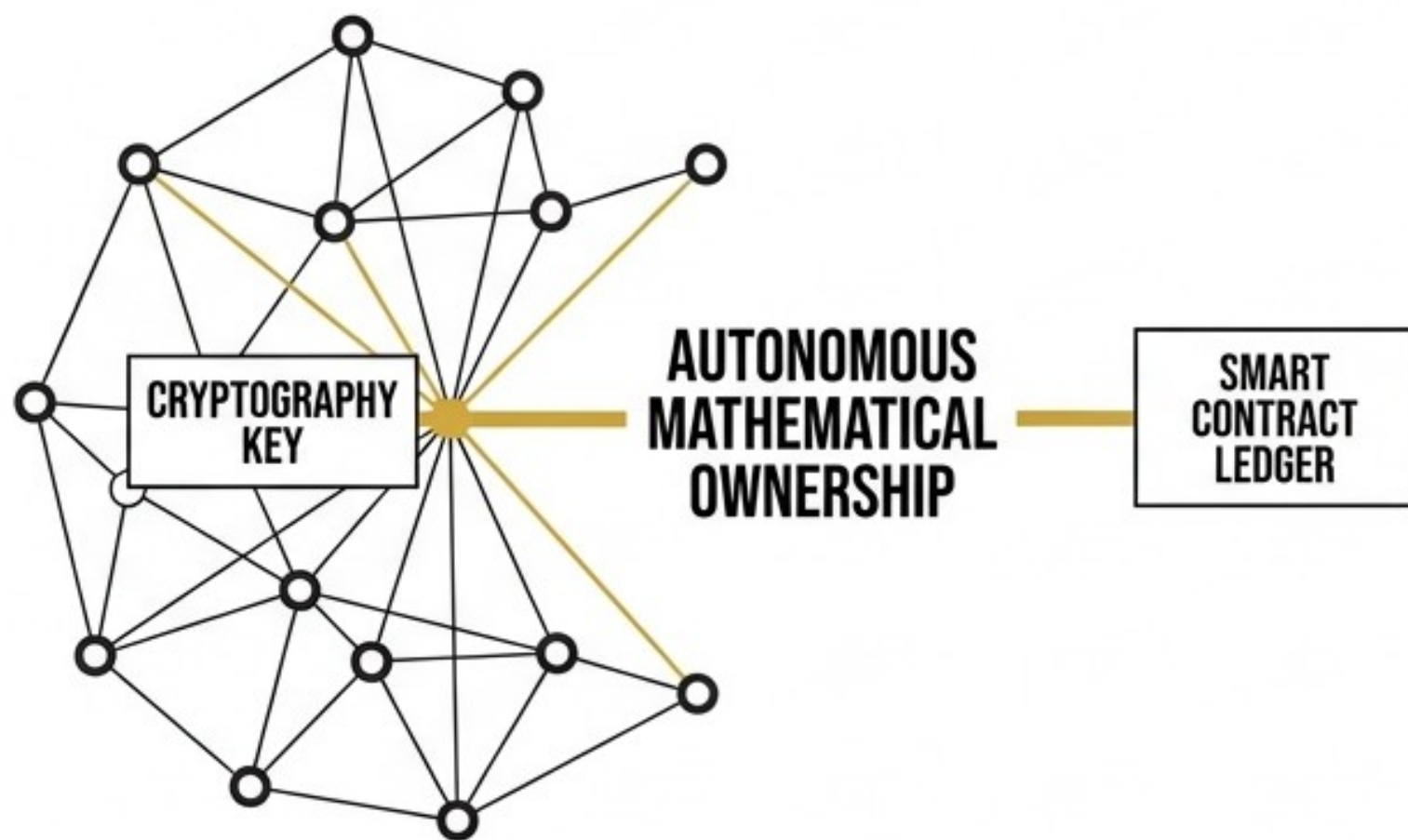
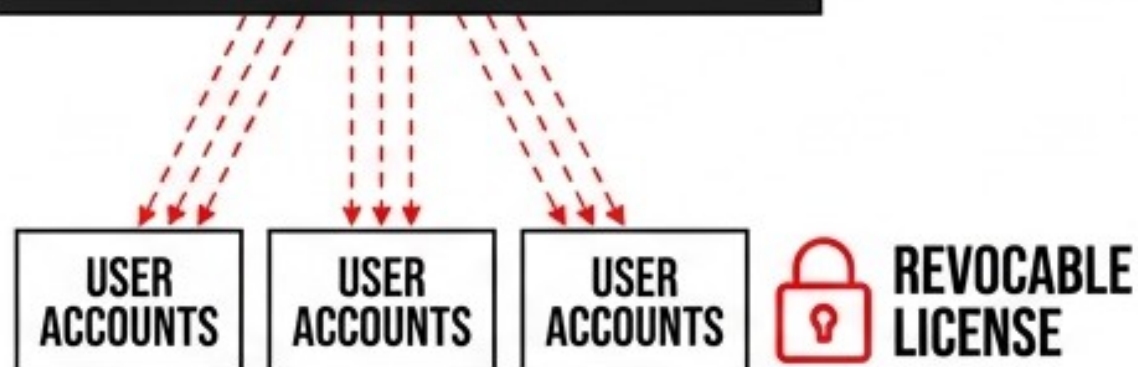


THE FUTURE OF DIGITAL OWNERSHIP

Exploring Ethereum Standards,
EIPs, and Token Models

CORPORATE PRIVATE DATABASE



THE PARADIGM SHIFT

Web2 Reality

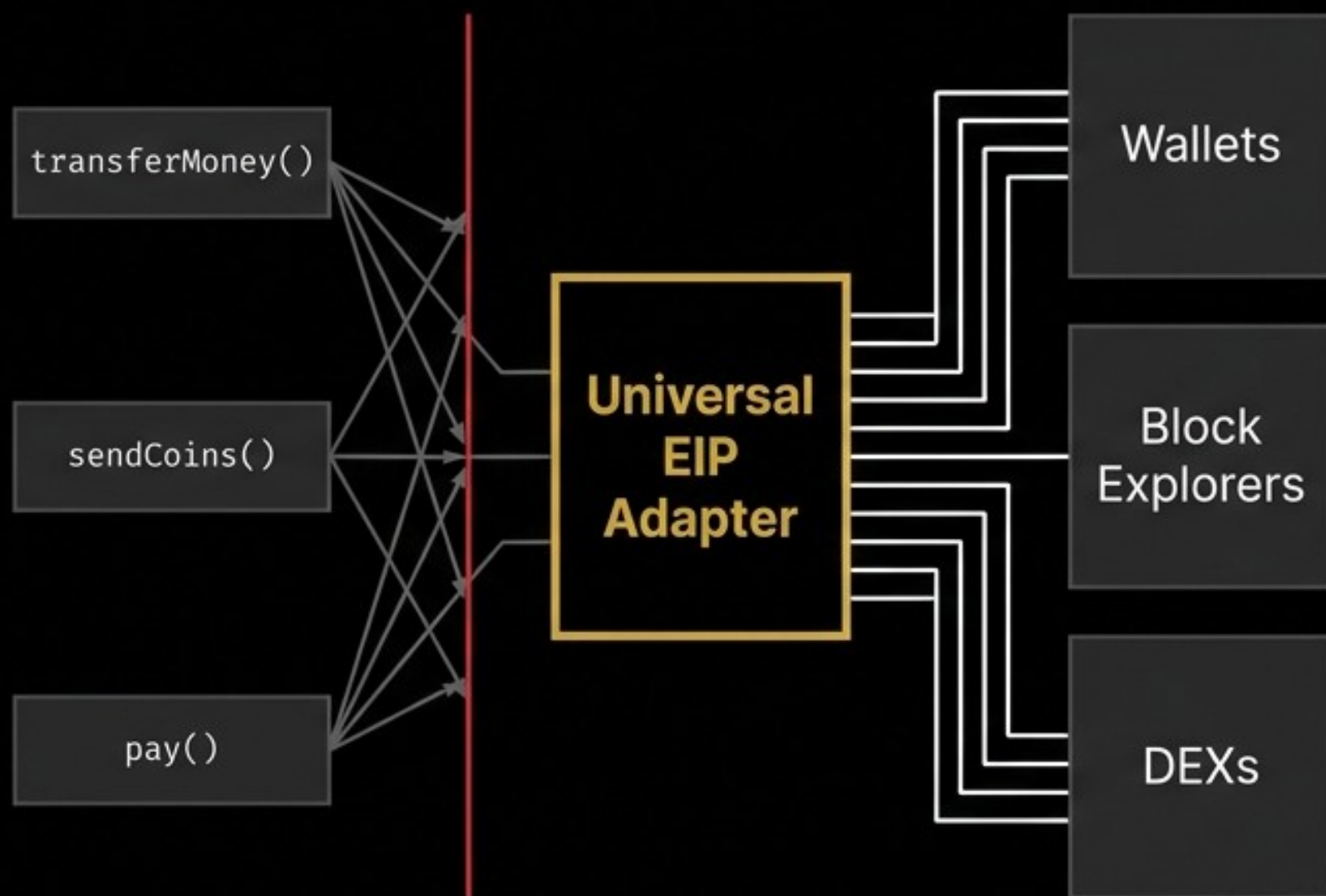
Purchasing digital assets like ebooks or game items grants only a revocable software license. Assets remain isolated inside corporate walled gardens.

Centralized Fragility

Accounts can be frozen unilaterally. Server shutdowns or provider bankruptcy instantly erase access to all assets.

The Blockchain Primitive

Smart contracts transform digital ownership from fragile paper agreements into autonomous, composable mathematical primitives on a public ledger.



THE IMPERATIVE OF COMPOSABILITY

The Dilemma

Without shared interface standards, every application must write custom integration code for every individual token.

Scaling Failure

Fragmented network effects destroy protocol interoperability and paralyze ecosystem growth before it can start.

EIP Solutions

Ethereum Improvement Proposals establish universal interface specifications. This creates instant composability—allowing autonomous smart contracts to interact seamlessly with any global asset without custom wrappers.

THE FOUNDATIONAL ASSET STANDARDS

1

ERC-20: Fungible Tokens

Identical, interchangeable units. Divisible by decimals.

Use Cases: Digital currencies, Stablecoins (USDC, DAI), Governance voting (UNI, AAVE).

2

ERC-721: Non-Fungible Tokens

Uniquely identifiable units (tokenId). Non-interchangeable.

Use Cases: Physical property certificates, Digital art provenance, ENS domain names.

3

ERC-1155: Multi-Token Standard

Hybrid standard consolidating fungible and non-fungible tokens into a single contract.

Use Cases: Mass-scale gaming inventories, Tiered event ticketing, Asset bundles.

```
mapping(address => uint256) private _balances);
```

Hex Address		Gold Integer

```
mapping(address => mapping(address => uint256))  
private _allowances);
```



INTERNAL ERC-20 MECHANICS

The Ledger Illusion

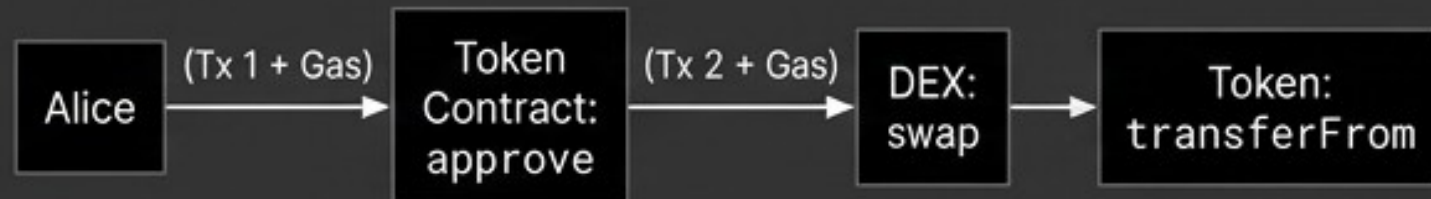
ERC-20 contracts are not vaults holding coins; they are decentralized spreadsheets governed by Solidity state mappings.

Six Mandatory Interfaces

Total compliance requires these universal functions:

- `totalSupply()`: Total circulating units.
- `balanceOf(account)`: Reads address state.
- `transfer(recipient, amount)`: Direct ledger update.
- `allowance(owner, spender)`: Checks third-party debit limits.
- `approve(spender, amount)`: Grants third-party debit permission.
- `transferFrom(sender, recipient, amount)`: Executes third-party withdrawal.

Path A (Traditional)



Path B (EIP-2612)



TWO-STEP ALLOWANCE VS EIP-2612

The Interaction Problem

Smart contracts cannot natively pull balances from a user's wallet without prior cryptographic authorization.

The Two-Step Pattern

Users must first execute an approve transaction, updating the `_allowances` mapping, before a protocol can call `transferFrom`.

Fatal Risk: Infinite approvals expose users to severe smart contract exploits.

Gasless Approvals via EIP-2612

Utilizes EIP-712 hashing for off-chain authorization. Users sign a payload (generating `v, r, s` cryptographic signatures) without paying gas. Applications bundle `permit()` and the execution function into one atomic transaction.

```
mapping(uint256 => address) private _owners;
```

tokenId: #8472

Wallet Hex Address

ERC-20: Address -> Integer | **ERC-721: Integer -> Address**

ABSOLUTE UNIQUENESS

The Nature of Non-Fungibility

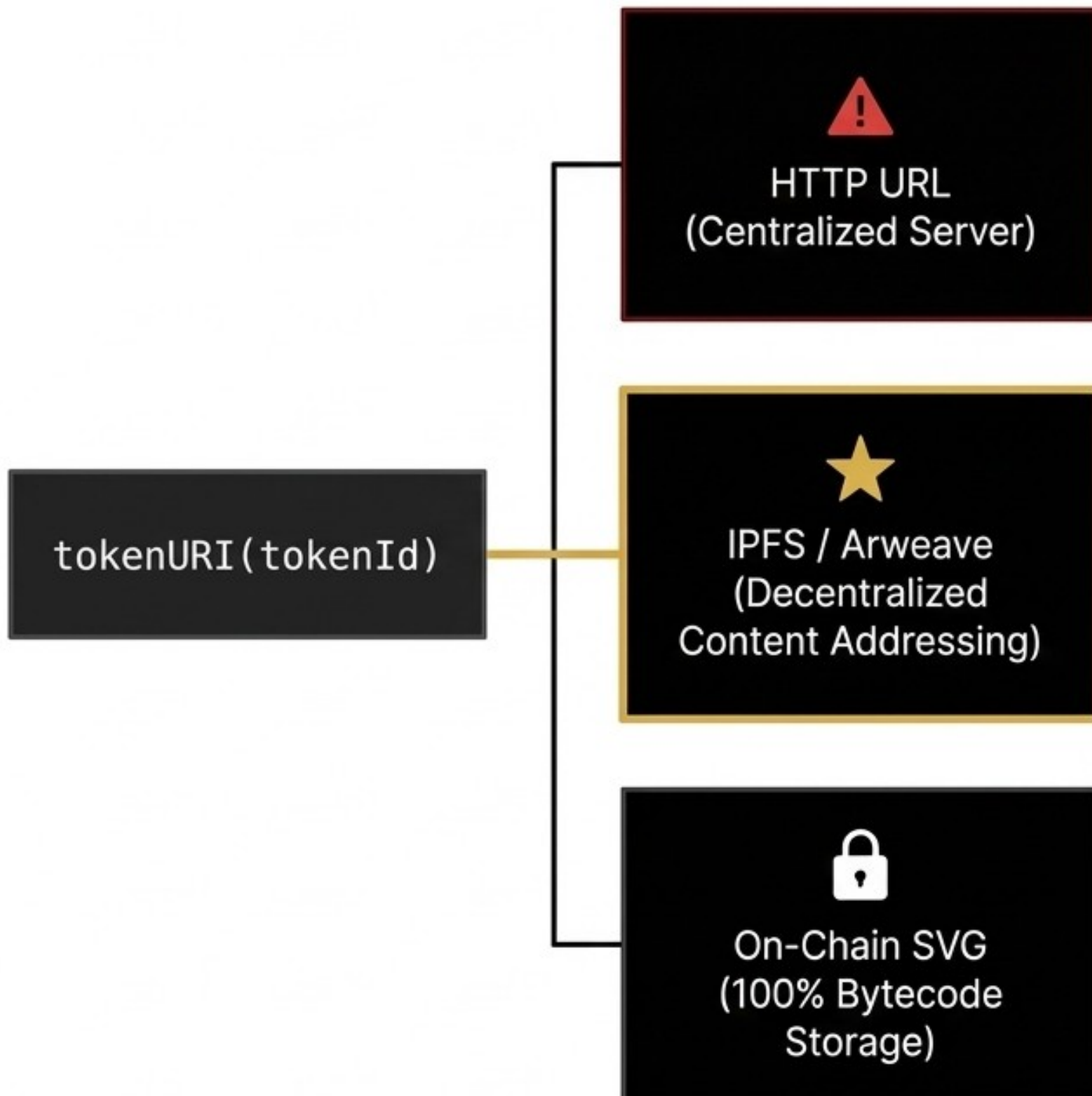
Designed for real-world and digital assets that cannot substitute one another, such as land deeds or digital identity.

Global Identification

Every unit is differentiated by a unique 256-bit unsigned integer ('tokenId'). Global identity is mathematically guaranteed by the exact combination: (ContractAddress, tokenId).

State Storage Structure

- **_owners**: Links a specific tokenId to its current legal address.
- **_balances**: Tracks the total quantity of NFTs held by an address.
- **_tokenApprovals**: Grants management rights over a singular tokenId.



TRANSFER SAFETY & METADATA

Black Hole Prevention

The `safeTransferFrom` function prevents permanent asset loss. It verifies the receiving contract implements the `IERC721Receiver.onERC721Received` selector. If unsupported, the transaction autonomously reverts.

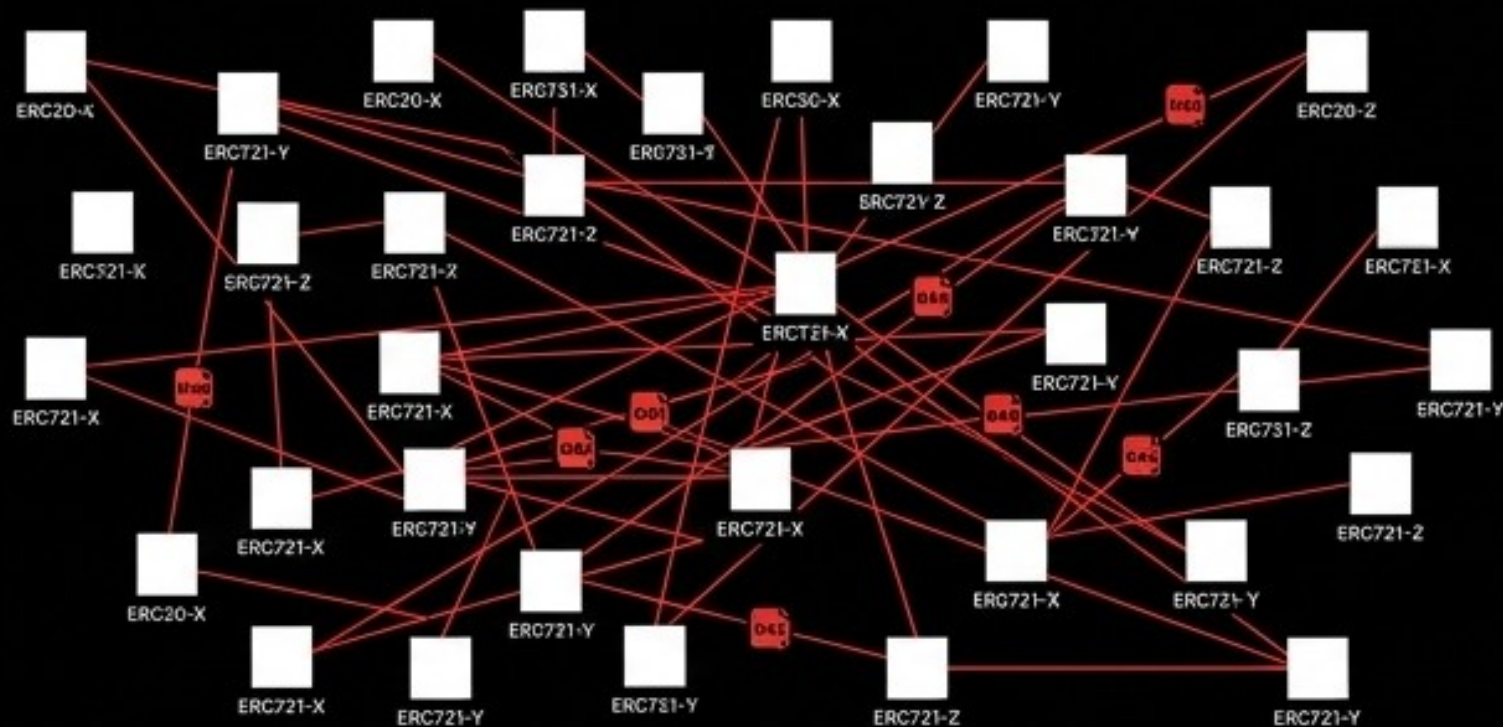
The Metadata Constraint

Storing high-resolution media directly on Ethereum L1 costs hundreds of thousands of dollars in gas. Contracts instead store lightweight string pointers accessed via `tokenURI`.

Storage Models

- **Centralized:** Private cloud servers (High risk of link rot).
- **Decentralized:** IPFS cryptographic hashes (Immutable industry standard).
- **On-Chain:** SVG visual code generated directly from smart contract bytes.

BEFORE



AFTER - ERC-1155

ERC-1155 Unified Contract

```
mapping(uint256 =>  
mapping(address => uint256))
```

safeBatchTransferFrom



Atomic
Batch
Transfer

ERC-1155 CONSOLIDATION

The On-Chain Gaming Problem

Complex protocols require thousands of disparate items (currencies, unique weapons, materials). Deploying and transacting across individual contracts is economically unviable.

Single State Consolidation

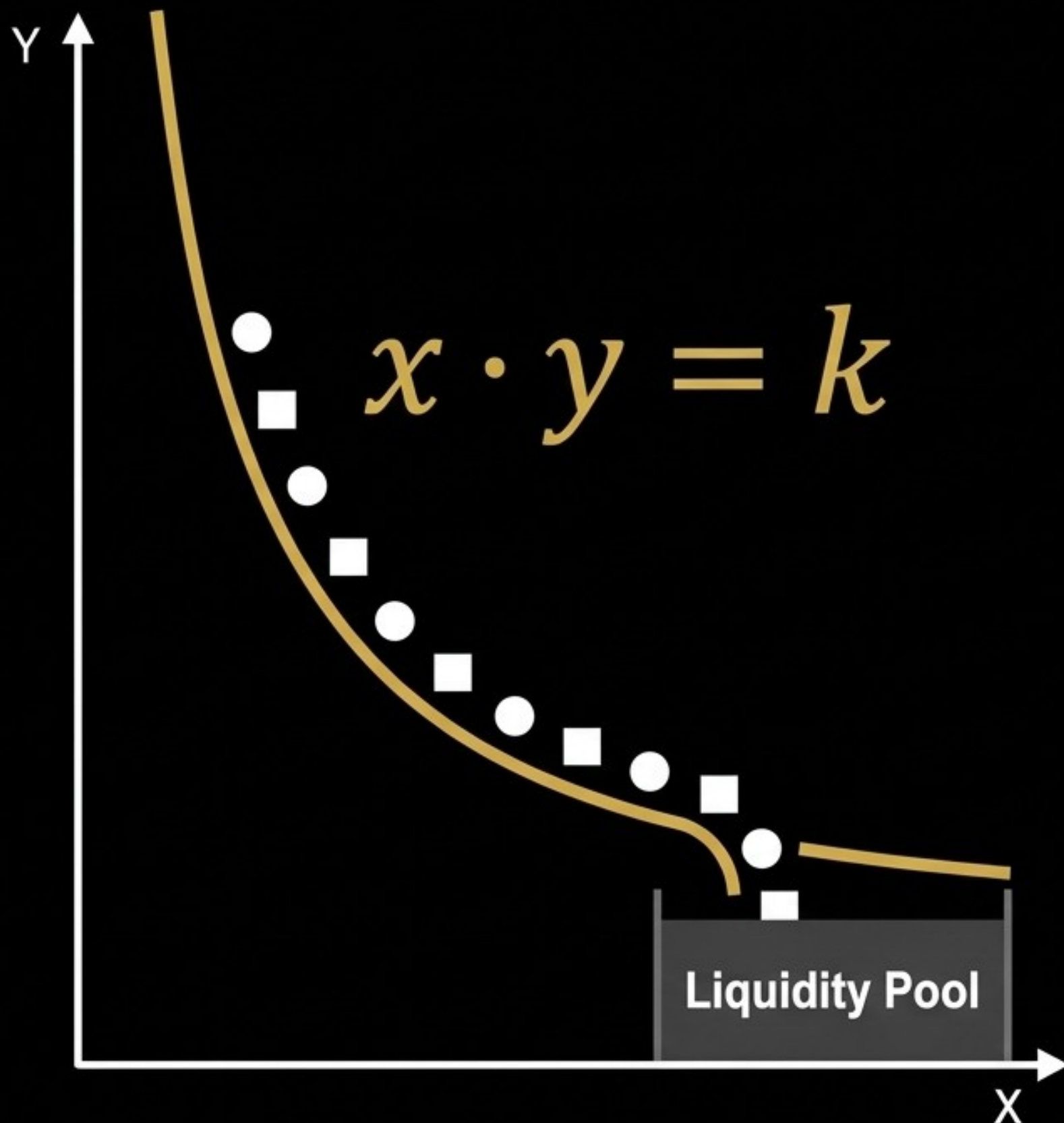
ERC-1155 manages fungible, non-fungible, and semi-fungible tokens within one unified nested mapping structure.

Atomic Batch Transfers

The `safeBatchTransferFrom()` function enables the transfer of dozens of different asset classes in a single atomic transaction. This slashes network gas **consumption by over 80%** compared to serial ERC-721 execution.

ARCHITECTURAL COMPARISON MATRIX

Dimension	ERC-20	ERC-721	ERC-1155
Asset Nature	Purely fungible, identical value.	Purely unique, 1-of-1 serialization.	Hybrid (fungible, semi-fungible, non-fungible).
State Storage Mapping	<code>address => uint256</code>	<code>uint256 => address</code>	<code>uint256 => (address => uint256)</code>
Execution Efficiency	Single transfer per call.	One transaction per NFT unit.	Native batch transfers & balance checks.
Deployment Overhead	One discrete contract per asset.	One discrete contract per collection.	One contract governs entire ecosystem.



BEYOND STANDARDIZATION

The Trading Imperative

Asset standardization successfully transforms ownership into composable math. The next architectural requirement is permissionless exchange without Wall Street intermediaries.

The L1 Order Book Failure

Traditional Central Limit Order Books (CLOBs) like NASDAQ fail entirely on Ethereum L1 due to exorbitant gas costs and block latency limits.

Next Module

Replacing centralized market makers with autonomous mathematical pricing curves ($x \cdot y = k$) and shared reserves: Automated Market Makers (AMMs) and Liquidity Pools.